HTTP Server

Serve a DuckDB connection through HTTP.

Install & load

```
SET allow_extension_repositories = 'allowed';
CREATE EXTENSION REPOSITORY IF NOT EXISTS query_farm WITH PREFIX 'https://duckdb-extensions.query.farm';
INSTALL httpserver FROM query_farm;  LOAD httpserver FROM query_farm;
```

01 At a glance

Start an HTTP listener inside a DuckDB process and query it from an HTTP client. Send SQL in a POST body, receive JSON results, and keep the process running for the lifetime of the service.

HTTP client → SQL request → DuckDB result

02 Server calls & HTTP contract

`httpserve_start(host, port, auth)`

Start the listener; with empty auth, only loopback hosts may be bound.

`httpserve_stop()`

Stop the listener; call start again to change settings.

`POST /`

Put SQL in the request body to execute it.

`GET /?q=…`

Pass URL-encoded SQL as a query parameter.

`GET /ping`

Unauthenticated liveness endpoint.

`default_format=JSONEachRow`

Newline-delimited JSON objects; numbers and booleans stay unquoted.

`SET httpserver_tls_cert / _tls_key`

PEM certificate and key files; the listener then speaks HTTPS.

`SET httpserver_allowed_origins`

Origins whose web pages may call the server; other browsers get 403.

03 Keep in mind

Choose the authentication mode

An auth value containing user:pass selects HTTP Basic; a shared key uses X-API-Key. An empty string serves loopback hosts only unless httpserver_allow_unauthenticated_remote is set. Beyond a trusted machine, configure TLS or a TLS proxy.

04 Start a local listener

MANUAL SETUP

```
SELECT httpserve_start(
  '127.0.0.1', 9999, 'replace-this-key'
);
```

Run in a dedicated DuckDB process and leave it open. Replace the example key.

05 Read from another DuckDB

REQUIRES SETUP

```
INSTALL httpfs;
LOAD httpfs;
CREATE SECRET quickref_http (
  TYPE HTTP,
  SCOPE 'http://127.0.0.1:9999/',
  EXTRA_HTTP_HEADERS MAP {
    'X-API-Key': 'replace-this-key'
  }
);
SELECT * FROM read_json_auto(
  'http://127.0.0.1:9999/?q=SELECT%2042'
  || '&default_format=JSONEachRow'
);
```

Use a second process after the listener starts; the matching key is required.

06 Stop from the serving process

MANUAL SETUP

```
SELECT httpserve_stop();
```

Call it from the serving connection, not from a served query; the listener also stops when its database closes.

07 In practice

Scope the DuckDB process

Clients can submit SQL using the served connection's capabilities. Configure DuckDB permissions and filesystem access for deployment; enable_external_access = false refuses httpserve_start. Read-only access alone is not a SQL sandbox. The server stops when its database closes.

Full docs & examples ↗ Source ↗